METACHRONAL CASCADE FLAPPER

Wave Parameter Reference Table

All maneuvers — both volumes — showing how each is executed through changes in metachronal wave parameters

Drone Prototype · Eight-Wing Staggered Cascade · 1 Hz Flap Frequency

HOW TO READ THIS TABLE

Each row represents one maneuver. The nine columns describe the metachronal wave parameter settings that produce that maneuver. Only the parameters that differ from the standard hover baseline need to be changed — all others remain at their default values. The rightmost column identifies the single most important change that defines each maneuver.

COLUMN	MEANING
Phase W1/W2/W3/W4	Degrees offset of each wing from the master clock. 0 deg = starts power stroke with master. 90 deg = starts 250ms after master at 1 Hz. 180 deg = starts 500ms after (opposite phase). Default metachronal = 0/90/180/270.
L vs R Asymmetry	Any difference between left-side and right-side wave parameters. Symmetric = both sides identical. Asymmetry generates roll, yaw, or lateral force depending on type.
Stroke Amplitude	Arc swept by each wing in degrees, measured from neutral position. Standard = 60 deg (plus or minus 60 = 120 deg total). Max = 65-70 deg. Reduced = 45-50 deg.
AoA Power / Recovery	Angle of attack held during power stroke and recovery stroke respectively. Standard power = 35 deg. Standard recovery = 5 deg (feathered). Both-stroke mode raises recovery AoA to 15-22 deg.
Sweep Schedule	Dynamic sweep position within each cycle. Standard = 30 deg fwd at mid-power-stroke, recovering aft during recovery. Full aft = stowed. Both-stroke uses symmetric fwd on both strokes.
Both-Stroke Mode	YES = recovery AoA raised to 15-22 deg to generate lift on both strokes. Required for inverted flight, vertical climb burst, and inverted maneuvers. Increases power consumption by 25-40%.

Key Wave Change	The single most important wave parameter modification that defines this maneuver. All other parameters may be at their standard values unless noted.

HOVER BASELINE — all maneuvers are described as deviations from this:

Phase: 0 / 180 / 0 / 180 deg · L vs R: fully symmetric · Amplitude: 60 deg all wings · AoA power: 35 deg · AoA recovery: 5 deg · Sweep: moderate fwd on power stroke, moderate aft on recovery · Both-stroke: NO

FUNDAMENTAL FLIGHT MODES								
MANEUVER	**PHASE W1/W2/ W3/W4**	**L vs R A SYMME TRY**	**STROK E AMP LITUDE**	**AoA P OWER**	**AoA R ECOV ERY**	**SWEEP SCHED ULE**	**BOT H- ST ROK E**	**PRIMARY WAVE CHANGE**
Stationary hover	0 / 180 / 0 / 180	Fully sy mmetric	60 deg all	35 deg	5 deg	Mod fwd power, mod aft recover y	No	Counter-ph ase pairs cancel body oscillation
Forward cruise	0 / 90 / 180 / 270	Fully sy mmetric	60 deg all	28-32 deg	3 deg	30 deg fwd at mid-stro ke	No	Stroke plane tilted 15-25 deg forward
Reverse flight	270 / 180 / 90 / 0	Fully sy mmetric	60 deg all	28-32 deg	3 deg	Revers ed sche dule	No	Wave direction flipped — W4 now leads
Vertical climb	0 / 45 / 90 / 135	Fully sy mmetric	65-70 deg all	40-42 deg	15-18 deg	Max forward	Yes	Phase com pressed toward in-phase; recovery AoA raised
Controlled descent	0 / 180 / 0 / 180	Fully sy mmetric	45-50 deg all	20 deg	20 deg	Modera te	No	Both strokes set to same drag AoA
Glide — wings fixed	ALL STO PPED	N/A	Wings locked	12 deg fixed	12 deg fixed	10 deg fwd fixed	N/A	Stroke stops; wings held at optimal glide AoA

TRANSLATIONAL MANEUVERS

MANEUVER	PHASE W1/W2/ W3/W4	L vs R A SYMME TRY	STROK E AMP LITUDE	AoA P OWER	AoA R ECOV ERY	SWEEP SCHED ULE	BOT H- ST ROK E	PRIMARY WAVE CHANGE
Lateral translation — crab	0 / 90 / 180 / 270	Stroke plane tilt asymmetry: L tilts right, R tilts left	60 deg all	28 deg	5 deg	Standard	No	Stroke plane tilt direction differs L vs R by 10-15 deg
Diagonal translation	0 / 90 / 180 / 270	Partial lateral tilt plus fwd tilt	60 deg all	28 deg	5 deg	Standard	No	Vector sum of fwd and lateral stroke plane tilts
Crosswind position hold	0 / 90 / 180 / 270	Dynamic — IMU computed	60 deg all	28 deg	5 deg	Standard	No	Crab angle updated each cycle by flight computer to cancel drift

ROTATIONAL MANEUVERS AND TURNS

MANEUVER	PHASE W1/W2/ W3/W4	L vs R A SYMME TRY	STROK E AMP LITUDE	AoA P OWER	AoA R ECOV ERY	SWEEP SCHED ULE	BOT H- ST ROK E	PRIMARY WAVE CHANGE
Spot yaw — pirouette	L leads R by 90-180 deg	L phase +90 deg; R phase -90 deg	60 deg symmetric	35 deg	5 deg	Standard	No	Opposing thrust vectors L vs R via phase; pure yaw couple
Banked turn — coordinated	0 / 90 / 180 / 270	Inside wing -20-30%; outside +20-30%	Differential L vs R	35 deg	5 deg	Standard	No	Amplitude differential for roll plus phase shift for yaw

Flat turn — skid	L leads R by 60-90 deg	Phase asymmetry only; amplitude equal	60 deg symmetric	35 deg	5 deg	Standard	No	Phase asymmetry generates yaw; no amplitude differential
Roll — aileron	0 / 90 / 180 / 270	L -30%; R +30% amplitude	Differential ±30%	Asymm ±5 deg	5 deg	Asymm — R more fwd	No	Three mechanisms: amplitude + AoA + sweep all differential
Pitch up	0 / 90 / 180 / 270	Fully symmetric	60 deg all	Rear +12 deg; front -8 deg	5 deg	Standard	No	Fore/aft AoA bias shifts centre of lift rearward
Pitch down	0 / 90 / 180 / 270	Fully symmetric	60 deg all	Front +12 deg; rear -8 deg	5 deg	Standard	No	Fore/aft AoA bias shifts centre of lift forward

FIXED-WING AEROBATIC MANEUVERS

MANEUVER	PHASE W1/W2/W3/W4	L vs R ASYMMETRY	STROKE AMPLITUDE	AoA POWER	AoA RECOVERY	SWEEP SCHEDULE	BOTH- STROKE	PRIMARY WAVE CHANGE
Loop — inside	0 / 90 / 180 / 270	Symmetric	60 deg all	Increasing through loop	5→18 inverted	Standard	Yes — inverted phase	Continuous nose-down pitch; both-stroke activated at >90 deg bank
Aileron roll	0 / 90 / 180 / 270	Max differential	Max differential	Compensated	18 inverted	Standard	Yes — inverted phase	Max roll input held 360 deg; both-stroke through inverted phase

Split-S	0 / 90 / 180 / 270	Half roll input	60 deg	Pull-through	18 inverted	Varies	Yes — brief	Half roll then continuous pull-through; exits forward
Immelmann turn	0 / 90 / 180 / 270	Symmetric climbing; half roll at top	Max climb then half roll	Max nose-up	18 at top	Max fwd climb	Yes — at apex	Full climb to apex; half roll exits in reversed heading
Hammerhead — stall turn	Hover mode at apex	Max yaw at apex	Max climb, then hover	Max climb then hover 35 deg	5 hover	Hover fwd at top	No	Climb to vertical; transition to hover; full yaw; dive
Knife-edge	0 / 90 / 180 / 270	L plane tilts up; R tilts down	60 deg	28 deg	5 deg	Standard	No	90 deg bank; stroke plane asymmetry provides vertical lift
Snap roll	0 / 90 / 180 / 270	Max roll differential	Max all wings	Max 42-45 deg all	Max 42-45 deg all	Standard	Yes — both strokes	Max roll PLUS max AoA all wings simultaneously — autorotation

ROTARY-WING MANEUVERS								
MANEUVER	**PHASE W1/W2/ W3/W4**	**L vs R A SYMMETRY**	**STROKE AMPLITUDE**	**AoA POWER**	**AoA RECOVERY**	**SWEEP SCHEDULE**	**BOTH- STROKE**	**PRIMARY WAVE CHANGE**

Pedal turn — in-place	L leads R 90 deg	Phase couple: L +90 R -90	Symmetric 60 deg	35 deg	5 deg	Standard	No	Pure phase couple — no amplitude or AoA change needed
Quick stop — deceleration	0 / 180 / 0 / 180	Symmetric	Max all	38-42 deg drag	38-42 deg drag	Aft tilt 30-40 deg	Yes	All wings drag mode both strokes; stroke plane tilted aft
Hover taxi	0 / 90 / 180 / 270	Symmetric	60 deg	28-32 deg	3 deg	3-8 deg fwd	No	Minimal fwd stroke plane tilt; IMU holds altitude
Nose-in hover	0 / 180 / 0 / 180	Symmetric	60 deg	35 deg	5 deg	Mod fwd	No	No wave change — FBW rotates pilot input to aircraft frame
Funnel — orbit	0 / 90 / 180 / 270	Symmetric	60 deg	28-32 deg	5 deg	Standard	No	Stroke plane tilt direction rotates continuously around orbit
Emergency glide	ALL STOPPED	N/A	Wings locked	12 deg fixed	12 deg fixed	10 deg fwd fixed	N/A	Wave stops; all servos hold fixed glide position
Inverted recovery	0 / 90 / 180 / 270	Max roll toward upright	60 deg	Standard	18 deg	Standard	Yes	Max roll + both-stroke + 25% throttle until upright

Wing fault recovery	Redistributed evenly	Mirror wing amplitude +15-20%	Compensated	35 deg	5 deg	Standard	No	Phase recalculated across remaining 7 wings automatically

QUADCOPTER FPV MANEUVERS

MANEUVER	PHASE W1/W2/W3/W4	L vs R ASYMMETRY	STROKE AMPLITUDE	AoA POWER	AoA RECOVERY	SWEEP SCHEDULE	BOTH- STROKE	PRIMARY WAVE CHANGE
Power loop	0 / 90 / 180 / 270	Symmetric	Max all	35 deg increasing	18 deg inverted	Standard	Yes — inverted phase	Max throttle + continuous pitch; both-stroke through inverted
Front flip	0 / 90 / 180 / 270	Symmetric	60 deg	35 deg	18 inverted	Standard	Yes — through inverted	Sustained pitch fwd through 360 deg; 3-5 seconds duration
Back flip	0 / 90 / 180 / 270	Symmetric	60 deg	35 deg	18 inverted	Standard	Yes — through inverted	Sustained pitch aft through 360 deg; reverse of front flip
Yaw spin	L leads R continuously	Phase: L +90 R -90 held	Symmetric	35 deg	5 deg	Standard	No	Full differential phase held; position hold via GPS

Inverted hover	0 / 90 / 180 / 270	Symmetric	60 deg	18-22 deg (lower surface)	18-22 deg (lower surface)	Standard	Yes — both strokes	Both-stroke AoA on lower surface; throttle +25-30%
Matty flip	0 / 90 / 180 / 270	Symmetric then half roll at top	Max	Nose-up arc max	18 at top	Fwd then over	Yes — at apex	Full nose-up arc over obstacle; half roll at apex
Split-S into hover	0 / 90 / 180 / 270	Half roll then pull-through	60 deg	Pull-through	18 inverted	Varies	Yes — brief	Pull-out throttle reduced to bleed speed into hover
Punch out	0 / 0 / 0 / 0	Fully symmetric	Max 70 deg	42 deg	18 deg	Max forward	Yes	All wings in-phase burst; both-stroke activated
Inverted yaw spin	L leads R continuously	Phase couple + inverted	Symmetric	18-22 deg inverted	18-22 deg inverted	Standard	Yes	Both-stroke inverted PLUS differential phase for yaw
Gap flying	0 / 90 / 180 / 270	Symmetric	60 deg	28 deg	5 deg	W3+W4 swept 60-80 deg aft	No	W3 and W4 swept aft reduces wingspan 35% for gap transit
Orbit / inverted orbit	0 / 90 / 180 / 270	Symmetric	60 deg	28 / 18 inverted	5 / 18 inverted	Standard	Yes — if inverted	Stroke plane direction rotates for orbit; both-stroke if inverted

BIRD-INSPIRED MANEUVERS								
MANEUVER	**PHASE W1/W2/ W3/W4**	**L vs R A SYMME TRY**	**STROK E AMP LITUDE**	**AoA P OWER**	**AoA R ECOV ERY**	**SWEEP SCHED ULE**	**BOT H- ST ROK E**	**PRIMARY WAVE CHANGE**
Pigeoning — slow turn	Yaw-domi nated — phase couple	Phase couple for body yaw	60 deg	35 deg	Both-st roke 18 deg	Standar d	Yes — low speed	Body yaws via phase couple; both-stroke maintains lift during yaw
Thermal soaring	STOPPE D or minimal	N/A	Wings locked	12 deg fixed	12 deg fixed	Fwd fixed	N/A	Wave stops; wings locked; gentle orbit in thermal column
Slope soaring	STOPPE D	N/A	Wings locked	12 deg fixed	12 deg fixed	Fwd fixed	N/A	Wave stops; altitude gained from ridge lift not power
Falcon stoop — dive	ALL STOWED	N/A	All wings stowed	N/A	N/A	Full aft — stow	N/A	All wings swept to full aft stow; gravity drives dive
Falcon stoop — pull-out	0 / 0 / 0 / 0	Symmetr ic	Max — all simul taneous	Max 45 deg	Max 45 deg	Max fwd sim ultaneo us	Yes	All 8 wings deploy simu ltaneously at full fwd sweep — peak LEV

Swift post-stall arrest	0 / 180 / 0 / 180	Symmetric	Max all	40-45 deg drag	40-45 deg drag	Aft 35 deg tilt	Yes	Both strokes at max drag AoA; stroke plane tilted aft
Undulating — bounding	Alternating: 0/90/180/270 then stopped	Symmetric	Alternating full/ stopped	35 deg flapping	5 deg flapping / locked glide	Alternating fwd/ fixed	No	Alternates between metachronal wave and glide every 0.8-1.5 sec
Perch landing	0 / 180 / 0 / 180 transitioning	Symmetric	Reducing to zero	35 deg then arrest	5 deg then drag	Fwd then standard	No	Swift arrest sequence then hover then vertical descent

BAT AND INSECT-INSPIRED MANEUVERS

MANEUVER	PHASE W1/W2/ W3/W4	L vs R ASYMMETRY	STROKE AMPLITUDE	AoA POWER	AoA RECOVERY	SWEEP SCHEDULE	BOTH- STROKE	PRIMARY WAVE CHANGE
Dragonfly 180-turn	Max yaw differential both sides	Max phase couple L vs R	Max all	35 deg	5 deg	Standard	No	Max phase couple PLUS in-phase for peak torque authority
Figure-8 hover	0 / 90 / 180 / 270	Symmetric	60 deg	35 deg	5 deg	Standard — increased deviation axis 20-25 deg at 2 Hz	No	Deviation amplitude increased; figure-8 tip path improves efficiency

Intercept dash	0 / 90 / 180 / 270	Oblique stroke plane — fwd + lateral vector	60 deg	28 deg	5 deg	30 deg fwd	No	Stroke plane tilted to vector sum of fwd and lateral components
Inertial flip	STOPPED	L swept full aft; R full fwd	L stowed; R max	N/A	N/A	Hard asymmetry: L full aft, R full fwd	N/A	Inertial torque from mass asymmetry — no aerodynamic force needed
Inverted ceiling landing	0 / 90 / 180 / 270 inverted	Symmetric then inertial flip	60 deg	18-22 inverted	18-22 inverted	Standard then R stow	Yes	Inverted hover then inertial flip to ceiling contact
Sharp bat turn	0 / 90 / 180 / 270	Inside: W3+W4 swept 60 deg aft; outside: full fwd	Differential	35 deg	5 deg	Hard asymmetry inside vs outside	No	Inertial asymmetry + phase couple + amplitude differential combined
Hawkmoth station-keep	0 / 180 / 0 / 180	Dynamic — updated each cycle	60 deg	35 deg	5 deg	Standard	No	Stroke plane tilt updated 1 Hz based on target position sensor
Moth evasion pattern	Random ±20 deg perturbation each wing each cycle	Random ±15% each cycle	Random ±15%	35 deg nominal	5 deg nominal	Standard	No	Stochastic perturbation of all parameters; unpredictable path

Escape jink	Single cycle: 270/180/90/0 then chosen heading	Max roll simultaneously	Max single cycle	Max	Max	Standard	No	Single reverse-wave cycle + max roll = velocity reversal in 2-3 sec

MANEUVERS UNIQUE TO THIS DESIGN

MANEUVER	PHASE W1/W2/W3/W4	L vs R ASYMMETRY	STROKE AMPLITUDE	AoA POWER	AoA RECOVERY	SWEEP SCHEDULE	BOTH- STROKE	PRIMARY WAVE CHANGE
Accordion flight	0 / 90 / 180 / 270 on active wings	Symmetric	Adapts as wings stow — +15-25 % per wing lost	35 deg	5 deg	Progressive stow W4→W3→W3→W4	No	Wing count cycles 8→6→4→6→8; phase redistributes around active wings
Standing wave hover	0 / 180 / 0 / 180 paired (W1+W3 vs W2+W4)	Symmetric	50 deg reduced	32 deg	5 deg	Moderate fwd	No	Paired resonant oscillation; smoothest possible hover; minimum oscillation
Asymmetric cascade	L: 0/90/180/270; R: 0/180/0/180	Different wave modes L vs R simultaneously	Equal both sides	Equal both sides	Equal both sides	Equal both sides	No	L side metachronal for thrust; R side counter-phase for smooth lift

Sequential burst launch	W1 deploys first; W2 +250ms; W3 +500ms; W4 +750ms	Symmetric	Max at each deployment	Max 42 deg	18 deg	0 deg — full fwd at deploy	Yes — at deployment	Each wing contributes its power stroke immediately on deployment
Phased drag braking	Sequential arrest: W1 first, then W2, W3, W4 at 250ms intervals	Symmetric	Max all	38-42 deg sequential	38-42 deg sequential	Aft sequential	Yes	Drag AoA activated per wing in sequence — smooth sustained deceleration
Wake surfing	Speed-adaptive: 90 deg at hover → 55 deg at 30 km/h	Symmetric	60 deg all	28-35 deg	5 deg	Standard	No	Phase offset updated by flight computer to track upstream vortex arrival
Mirror wave lateral	L: 0/90/180/270; R: 270/180/90/0	Exact temporal mirror L vs R	Symmetric baseline; biased for direction	35 deg	5 deg	Standard	No	Mirror timing creates 1 Hz lateral lift oscillation; bias gives translation

SUMMARY — HOW THE WAVE CONTROLS EVERYTHING

Every maneuver in this table is achieved by modifying one or more of seven parameters within the metachronal wave generator. No mechanical reconfiguration is ever required between flight modes. The following summarises how each parameter class maps to flight outcomes:

PHASE OFFSET (W1/W2/W3/W4 degrees)	The most powerful control parameter. Changes the wave travel speed and direction. Metachronal (0/90/180/270) = efficient forward cruise. Counter-phase (0/180/0/180) = smooth hover. In-phase (0/0/0/0) = maximum lift burst. Reversed (270/180/90/0) = reverse flight. Differential L vs R = yaw torque.
L vs R ASYMMETRY (phase, amplitude, or AoA)	Any left-right difference generates a moment. Phase asymmetry = yaw. Amplitude asymmetry = roll. AoA asymmetry = roll with pitch coupling. Sweep asymmetry = roll with inertial component at low speed. Three independent asymmetry mechanisms per axis = triple redundancy.
STROKE AMPLITUDE (degrees of arc)	Sets total lift and thrust magnitude. Symmetric changes = throttle. Asymmetric changes = roll control. Maximum amplitude = punch-out/burst climb. Reduced amplitude = gentle descent or loiter. Wings stowed = full aft sweep.
AoA POWER / RECOVERY (angle of attack each stroke)	Power AoA sets lift on the power stroke. Recovery AoA sets drag on return. Low recovery AoA (5 deg) = efficient feathered recovery. High recovery AoA (18-22 deg) = both-stroke lift for inverted/climb. Equal power and recovery AoA = drag mode for braking or descent. Fore/aft differential = pitch control.
SWEEP SCHEDULE (within each cycle)	Forward sweep during power stroke = maximum span, maximum CL. Aft sweep during recovery = minimum drag, efficient return. Full aft = stow. Asymmetric sweep = roll + inertial reorientation. Speed-adaptive phase offset = wake surfing / inter-wing vortex capture.
BOTH-STROKE MODE (recovery AoA raised)	Activating both-stroke mode (recovery AoA raised to 15-22 deg) enables: inverted flight, vertical climb burst, inverted hover, bat-style landing, and any maneuver requiring lift through 180 degrees of attitude. Costs 25-40% additional power.
WAVE DIRECTION (front-to-back vs back-to-front)	Traveling wave front-to-back = forward thrust. Back-to-front = reverse thrust. Pure software change — no mechanical reversal. Combined with amplitude and AoA control, wave direction sets the complete thrust vector in three-dimensional space.

COMPLETE CONTROL PRINCIPLE: The metachronal wave is a seven-dimensional control surface. Every combination of phase, asymmetry, amplitude, AoA power, AoA recovery, sweep schedule, and wave direction produces a distinct flight behaviour. The 55+ maneuvers in this table represent specific named points within that seven-dimensional space. The transitions between them are smooth — the flight computer interpolates between parameter sets over 2-3 wing cycles, producing fluid maneuver entries and exits with no discontinuities.

End of Document